

Mathematics 629 — Homework 7

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Problem 1

Let $X_1 = [0, 1]$, with $\mathcal{M}_1 = \mathcal{B}_{[0,1]}$ and with μ_1 the Lebesgue measure, $X_2 = [0, 1]$ with $\mathcal{M}_2 = \mathcal{B}_{[0,1]}$ and μ_2 being counting measure. Let $\mu = \mu_1 \times \mu_2$. Consider the diagonal $D = \{(x, x) : 0 \leq x \leq 1\}$.

- (i) Use the definition of μ to evaluate $\int \mathbf{1}_D d\mu$.

Answer. By definition of product measure, $\int \mathbf{1}_D d\mu = \int \mu_2(sl_{x_1}) d\mu_1(x_1)$ where $sl_{x_1} = x_1$. Then $\int \mathbf{1}_D d\mu = \int_{[0,1]} 1 d\mu_1 = 1$. $\square$

- (ii) Show that $\int \int \mathbf{1}_D d\mu_1 d\mu_2$ and $\int \int \mathbf{1}_D d\mu_2 d\mu_1$ have different values. Why can Tonelli's theorem not be applied?

Answer. In (i), integrating with respect to μ_2 first showed $\int \int \mathbf{1}_D d\mu_2 d\mu_1 = 1$. Let $sl_{x_2} = x_2$. Then similarly,

$$\int \int \mathbf{1}_D d\mu_1 d\mu_2 = \int \mu_1(sl_{x_2}) d\mu_2(x_2) = 0 * (sl_{x_2}) = 0.$$

Clearly order of integration matters in this case. Since Tonelli guarantees equivalence regardless of order of integration, the theorem cannot be applied here because $(X_2, \mathcal{M}_2, \mu_2)$ is not sigma finite ($\mu_2([0, 1])$ is uncountable). $\square$

Problem 2

Consider, for $x \in (0, 1) \times (0, 1)$,

$$f(x_1, x_2) = \frac{x_1^2 - x_2^2}{(x_1^2 + x_2^2)^2} = \frac{\partial^2}{\partial x_1 \partial x_2} \arctan \frac{x_1}{x_2}.$$

- (i) Verify that

$$\int_0^1 \int_0^1 \frac{x_1^2 - x_2^2}{(x_1^2 + x_2^2)^2} dx_2 dx_1 = \frac{\pi}{4}, \quad \int_0^1 \int_0^1 \frac{x_1^2 - x_2^2}{(x_1^2 + x_2^2)^2} dx_1 dx_2 = -\frac{\pi}{4}.$$

Answer. (a) First observe that since $\frac{\partial^2}{\partial x_1 \partial x_2} \arctan \frac{x_1}{x_2} = f$ then

$$\frac{\partial}{\partial x_2} \arctan \frac{x_1}{x_2} = \frac{1}{1 + (x_1/x_2^2)} * \left(\frac{-x_1}{x_2^2} \right) = \frac{-x_1}{x_1^2 + x_2^2}.$$

In other words,

$$\int_0^1 \int_0^1 f dx_2 dx_1 = \int_0^1 \frac{1}{1 + x_1^2} dx_1 = \arctan(x_1)|_0^1 = \frac{\pi}{4}.$$

Similarly, integrating in the other order shows

$$\frac{\partial}{\partial x_1} \frac{-x_1}{x_1^2 + x_2^2} = f$$

so then by keeping x_1 fixed we get that

$$\int_0^1 \int_0^1 f dx_1 dx_2 = \int_0^1 \frac{-1}{1 + x_2^2} dx_2 = -\arctan(x_2)|_0^1 = -\frac{\pi}{4}.$$

□

(ii) Let $T = \{x : 0 < x_2 < x_1 < 1\}$, let m be Lebesgue measure in $(0, 1)^2$ and show that

$$\int_T f dm = \infty.$$

Answer. Observe that using $\frac{\partial}{\partial x_2} \left(\frac{x_2}{x_1^2 + x_2^2} \right) = f$ then

$$\int_0^{x_1} = \left[\frac{x_2}{x_1^2 + x_2^2} \right]_{x_2=0}^{x_2=x_1} = \frac{1}{2x_1}.$$

Then integrating over x_1 shows

$$\int_T f dm = \int_0^1 \frac{1}{2x_1} dx_1 = \frac{1}{2} [\ln x_1]_0^1 = \infty$$

□

(iii) Why can Fubini's theorem not be applied in (i)?

Answer. Fubini's theorem requires that $f \in L^1((0, 1)^2)$ but we have shown this isn't true in part (ii). This is why part (i) yields integrals of different values without contradicting Fubini's Theorem. □

Problem 3

Let $(X_1, \mathcal{M}_1, \mu_1)$ be a measure space and $X_2 = \mathbb{N}$ with $\mathcal{M}_2 = \mathcal{P}(\mathbb{N})$ and counting measure μ_2 . Interpret Tonelli's and Fubini's theorem in this situation and show that one does not need the assumption of σ -finiteness of μ_1 in this case.

Answer. Let $f : X \times \mathbb{N} \rightarrow \mathbb{R}$ and let $f(n(x_1)) = f(x_1, n)$ for $x_1 \in X_1$ and $n \in \mathbb{N}$. Recall that counting measure is a sum so

$$\int_{\mathbb{N}} f(x_1, n) d\mu_2 = \sum_{n=1}^{\infty} f(n(x_1)).$$

By Tonelli's theorem,

$$\int_{X_1} \sum_{n=1}^{\infty} f_n(x_1) d\mu_1 = \sum_{n=1}^{\infty} \int_{X_1} f(n)x_1 d\mu_1.$$

However this result follows as a result of MCT. If we let $S_n = \sum_{n=1}^N f(n(x_1))$ be the partial sum up to some N then by MCT

$$\int_{X_1} \sum_{n=1}^{\infty} f_n d\mu_1 = \lim_{N \rightarrow \infty} \int_{X_1} \sum_{n=1}^N f_n d\mu_1 = \lim_{N \rightarrow \infty} \sum_{n=1}^N \int_{X_1} f_n d\mu_1 = \sum_{n=1}^{\infty} \int_{X_1} f_n d\mu_1.$$

This proof doesn't rely on the σ -finiteness of μ_1 and furthermore, to show $\sum_N \int |f_n| d\mu_1 < \infty$ apply Tonelli's theorem to f_n^+ and f_n^- . In both cases, σ -finiteness of μ_1 is not needed since sums over $\mathbb{N}$ are always well defined. $\square$

Problem 4

Let $(X, \mathcal{M}, \mu)$ be a σ -finite measure space. Let f be a nonnegative measurable function on X and

$$U_f = \{(x, y) \in X \times [0, \infty] : 0 \leq y \leq f(x)\}.$$

Show that U_f is $\mathcal{M} \times \mathcal{B}_{\mathbb{R}}$ measurable and $\mu \times m(U_f) = \int f d\mu$. Show that the Lebesgue measure (in $\mathbb{R}^2$) of the graph of f (i.e., $\{(x, f(x)) : x \in X\}$) is zero.

Answer. Since we have σ -finiteness then let $sl_x = sl_x([0, \infty]) = \{y : (x, y) \in U_f\}$. Then integrating over sl_x observe that

$$\int_0^{\infty} \mathbf{1}_{U_f}(sl_x) dm(y) = f(x)$$

since

$$\mu \times m(U_f) = \int_0^{\infty} \int_X \mathbf{1}_{U_f} d(\mu \times m).$$

In other words, $\mu \times m(U_f) = \int_X f(x) d\mu(x) = \int f d\mu$. Let $G_f = \{(x, f(x)) : x \in X\}$ be the graph of f . Note that $sl_y(G_f) = \{f(x)\}$ which as a singleton has $m(sl_y) = 0$. Thus

$$\mu \times m(G_f) = \int_X m(sl_y) d\mu(x) = \int_X 0 d\mu = 0.$$

So the measure of the graph of f is 0. $\square$

Problem 5

Let

$$A_1 := \{(x, y, z) \in \mathbb{R}^3 : x^2 + z^2 \leq 1\}, \quad A_2 := \{(x, y, z) \in \mathbb{R}^3 : y^2 + z^2 \leq 1\}$$

and compute the Lebesgue measure of $A_1 \cap A_2$.

Answer. Define $sl_{xy}(E) \in \mathbb{R}^{\neq}$ as the generalized rectangle for fixed z given some $E \in A_1 \cap A_2$ and $sl_x(E) \in \mathbb{R}$ as the generalized rectangle for fixed y and z . Note that $m = m_1 \times m_2 \times m_3$ where m_1, m_2, m_3 are Lebesgue measure. Since Lebesgue measure is σ -finite, by the generalized Cavalieri principle we can compute

$$m(A_1 \cap A_2) = \int m_1 \times m_2(sl_{xy}) dm_3(z) \tag{1}$$

$$= \int \int m_2(sl_x(E)) dm_1(x) dm_3(z) \tag{2}$$

$$= \int_{-1}^1 \int_{\sqrt{1-z^2}}^{\sqrt{1-z^2}} 2\sqrt{1-z^2} dm_1(x) dm_3(z) \tag{3}$$

$$= \int_{-1}^1 4(1-z^2) dm_3(z) = \frac{16}{3}. \tag{4}$$

□